Southern Cross University

Second Part/Chapter of the Autonomic Nervous System (ANS) Pharmacology

Introductory Pharmacology
PHAR2001

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Objectives for second chapter on the ANS drugs

Describe the actions, interactions and side effects of the following adrenergic agonists:

- Adrenaline (non-selective agonist)
- Phenylephrine (an alpha 1 selective agonist)
- Dobutamine (a beta 1 adrenergic agonist)
- Salbutamol (a beta 2 adrenergic agonist)
- Dopamine

These medications will be taken again at a later time. It is most important right now to understand why the actions and side effects occur

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Sympathomimetics (Drugs that mimic the sympathetic nervous system (SNS)

Lesson 1:

Understanding Anaphylaxis and the Role of Adrenaline

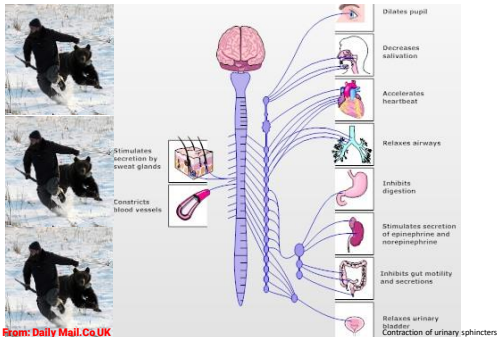
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Sympathomimetic drugs

- Sympathomimetics
- Adrenergic agonists
- Alpha agonist
- Beta agonist
- Adrenergic receptor stimulators

These terms are often used interchangeably and therefore should all be known as drugs that increase the sympathetic response

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Non-selective sympathomimetic

- **Specific drugs:**
  - Adrenaline (know this drug)
  - Noradrenaline
  - Metaraminol
  - Ephedrine
  - Dopamine (know this drug)
  - Pseudoephedrine (Know; combined with antihistamines and analgesics)
- **These have limited use due to their effect on a wide range of organs, but are still used for treatment of hypotension and anaphylactic shock.**
  - After intravenous administration of adrenaline, an Alpha-adrenergic antagonist should be administered locally to offset the vasoconstricting effects of adrenaline.

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Hypersensitivity reactions

Type I (Anaphylaxis, angioedema)

Type II (Example: nephritis)

Type III (Example: vasculitis

Type IV (Example: contact dermatitis)

NOT AN ALLERGY!

Only a Type I hypersensitivity reaction is a true allergy!

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Hypersensitivity reactions

Type I (Anaphylaxis, angioedema)

Allergic rhinitis is a
type I hypersensitivity

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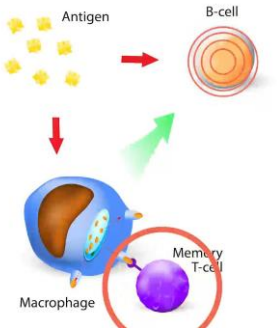


Allergens

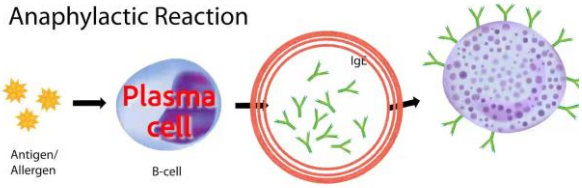
(Non-specific: antigen)

* Bee sting
* Peanuts
* Shell fish
* Fish
* Penicillin
* Other drugs...

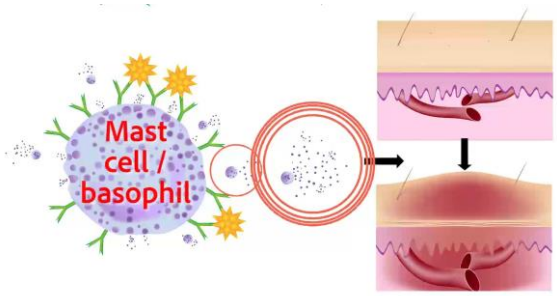
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Antigen
B-cell
Macrophage

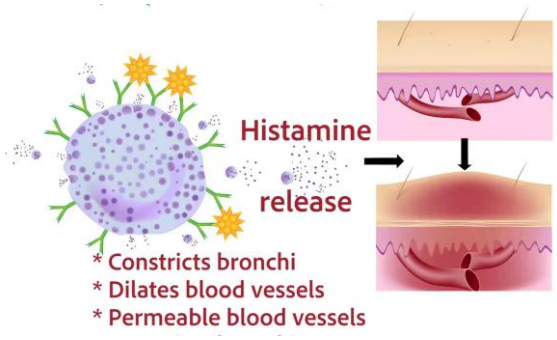
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Anaphylactic Reaction
Plasma cell
Antigen/
Allergen
B-cell

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Mast cell / basophil

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Histamine
release
* Constricts bronchi
* Dilates blood vessels
* Permeable blood vessels

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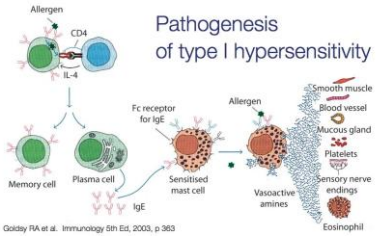

Angioedema is seen in >50% of cases
Angioedema can be a reaction on its own
* Difficulty breathing
* Low BP
* Possible Angioedema
* Oedema

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Epipen is an autoinjector or adrenaline (epinephrine)

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Anaphylaxis (Anaphylactic shock)



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Sympathomimetics
Lesson 2:
Non-selective adrenergic agonists other than adrenaline

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Non-selective sympathomimetic

- **Specific drugs:**
  - **Adrenaline (know this drug)**
  - **Noradrenaline**
  - **Metaraminol**
  - **Ephedrine**
  - **Dopamine (know this drug)**
  - **Pseudoephedrine (Know; combined with antihistamines and analgesics)**
- **These have limited use due to their effect on a wide range of organs, but are still used for treatment of hypotension and anaphylactic shock.**
  - **After intravenous administration of adrenaline, an Alpha-adrenergic antagonist should be administered locally to offset the vasoconstricting effects of adrenaline.**

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‘Selective’ adrenergic receptor stimulation

Subtypes of adrenergic receptors

- Alpha
  - Alpha 1
  - Alpha 2
- Beta
  - Beta 1
  - Beta2

While drugs that target these ‘selective’ receptors have more selectivity, they still result in some stimulation to the other receptors, especially in high doses.

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Sympathomimetics
Lesson 3:
Selective alpha adrenergic agonists

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‘Selective’ adrenergic receptor stimulation

Subtypes of adrenergic receptors

- Alpha
  - Alpha 1
  - Alpha 2
- Beta
  - Beta 1
  - Beta2

While drugs that target these ‘selective’ receptors have more selectivity, they still result in some stimulation to the other receptors, especially in high doses.

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Alpha 1 receptors

Location of receptors:

- 'Peripheral' blood vessels (including those in the nasal passages; large muscles do not have this constriction to ensure they get enough blood to move quickly)
- Smooth muscles of the GIT (less effect than Beta 1 stimulation)
- Sphincters of the bladder
- Radial muscles of the eye

☞ ***Given the location of the receptors, hypothesise the uses and side effects of Alpha 1 sympathomimetics***

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Alpha 1 agonists: side effects

Any sympathetic nervous system actions can be increased, but the most common adverse effects include:

- Hypertension
- Constipation
- Urinary retention

- (eye pain is possible)



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Actions: Alpha 1 agonists

- Nasal passages and eyes:
  - The constrictive effects of Alpha 1 adrenergic agonists can counteract the vasodilation that leads to nasal congestion and the 'red eye'.
- Systemically:
  - Increase in blood pressure (through a constriction of blood vessels).
- Dilate the pupils
  - Allows as much light in as possible but can also lead to eye pain when too much light is being allowed in.
- In minor surgery
  - Local administration constricts blood vessels for less bleeding and decreased systemic absorption of local anaesthesia

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Alpha 1 agonists

(Most end in -ine, like epinephrine but that is not a formal suffix)

Phenylephrine (know this one; used primarily as a decongestant, as an agent to dilate the pupils, and to increase blood pressure.)
Midodrine (oral decongestant)
Naphazoline (intranasal decongestant)
Oxymetazoline (know this drug; intranasal decongestant)
Xylometazoline (intranasal decongestant)

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Alpha 2 agonists

This is the exception to the rule so ignore for now.

Explanation: the Alpha 2 receptor exists to make sure that the adrenaline response does not result in so much stimulation that it can do harm. It is a 'negative feedback' type of receptor and does exactly opposite to that which one would expect.

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Sympathomimetics
Lesson 4:
Selective beta adrenergic agonists

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Beta 1 adrenergic receptors

Location

- Heart
- Git
- Kidneys

☞Given the location of the receptors, hypothesise the uses and side effects of Beta 1 agonists (selective Beta 1 sympathomimetics)

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Actions: Beta 1 adrenergic agonists

- Positive ***chronotropic*** effects (increased heart rate)
- Positive ***inotropic*** effect (increase in the contractile force of the heart)
  - *Know these terms – chrono=time (beats/min) inos=fibre (how much stretch*
- Therefore these drugs are useful in circulatory shock, hypotension and cardiac arrest
- Adverse effects
  - GIT – constipation
  - Kidneys – decreased blood flow = increased systemic blood pressure via the renin angiotensin aldosterone mechanism
- Example: Dobutamine (know this drug; used to treat acute but potentially reversible heart failure)

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Beta 2 receptors

Location

- Lung
- Mast cells
- Skeletal muscle

☞Given the location of the receptors, hypothesise the uses and side effects of Beta 2 agonists (selective Beta 2 sympathomimetics)

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Actions: Beta 2 agonists

- Bronchodilation
- Mast cell stabilisation
- (Side effect: often associated with muscle tremor)

- Used in chronic obstructive pulmonary disease, acute asthma and premature labour
- Eformoterol, indacaterol, salmeterol
- Salbutamol (know this drug; used for the relief of bronchospasm in conditions such as asthma and COPD

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References

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